

Recommendation stream by grain: OnlineRFPerm, RFPerm/CFPerm, and FSDS

Y (outcome). One row is one order. $Y \in \{0, 1\}$ is **conversion** on that order (purchase / not). Y is the response only. It is never a column of X , never a ranking key, never a subset key.

X (covariates). X is the serving table of that order, sliced the way the log is written. Y is not in any slice.

grain	X columns	what the row is
order	amount, hour, n_items, channel	this order
merchant	merchant_cat, merchant_gmv, n_skus	the merchant on this order
user	user_tenure, user_hist_freq	the user on this order
all	the nine columns concatenated	anti-pattern; shown as a check

Region (south / north) is a merchant attribute, not a feature and not Y . The batch index T is a time label (D_{ref} vs the new batch), not a treatment.

Setup. $n_{\text{ref}}=400$, $n_{\text{new}}=120$, six batches, labeled onset $t=2$. After onset, only **south** merchants are shifted. Covariate / both plant amount $\succ$ merchant_gmv $\succ$ channel in $P(X)$. Concept plants amount in $P(Y | X)$ only. User columns never walk.

Table 1: OnlineRFPerm (Algorithm 1) — first rejection, delay, FAR.

Table 2: FSDS / RFPerm ΔMSE / CFPerm φ -VIMP — Kendall- τ vs planted.

Table 3: south vs north own-ref. **MMD, CMean, and PO-risk together.**

Table 1: WHEN. Frozen RF on D_{ref} . $T_t = \text{MSE}_t - E_{\text{ref}}$. Delay = first rejection — onset. FAR = mark before onset.

	kind	grain	ADDIS	rank- p	SAFFRON	hop	last T	last MMD
	covariate	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.067	0.0952
	covariate	merchant	5 ($d=3$)	2 ($d=0$)	5 ($d=3$)	miss	0.066	0.0645
	covariate	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	0.0569	0.000627
	covariate	all	FAR@0	FAR@0	FAR@0	miss	0.0531	0.0773
	concept	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.113	−0.000532
	concept	merchant	miss	2 ($d=0$)	miss	miss	0.0188	−0.00488
	concept	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	0.00947	0.000627
	concept	all	FAR@0	FAR@0	FAR@0	miss	0.11	−0.00121
	both	order	2 ($d=0$)	2 ($d=0$)	2 ($d=0$)	miss	0.157	0.0952
	both	merchant	miss	2 ($d=0$)	miss	miss	−0.0187	0.0645
	both	user	2 ($d=0$)	FAR@0	2 ($d=0$)	miss	−0.000541	0.000627
	both	all	FAR@0	FAR@0	FAR@0	miss	0.17	0.0773

Table 2: WHICH columns. Ranking recovery vs planted magnitude. User grain has no planted X columns — the correct negative control.

kind	grain	planted in X	FSDS recovered	τ_{FSDS}	τ_{RFPPerm}	τ_{CFPerm}
covariate	order	amount, channel	amount, channel	0.913	0.548	-0.183
covariate	merchant	merchant_gmv	merchant_gmv	0.816	-0.816	-0.816
covariate	user	—	—	—	—	—
covariate	all	amount, channel, merchant_gmv	amount, channel, merchant_gmv	0.691	0.182	-0.182
concept	order	amount	amount	0.236	0.236	0.707
concept	merchant	—	—	—	—	—
concept	user	—	—	—	—	—
concept	all	amount	amount	0.354	0.354	0.471
both	order	amount, channel	amount, channel	0.913	-0.183	0.913
both	merchant	merchant_gmv	merchant_gmv	0.816	0	-0.816
both	user	—	—	—	—	—
both	all	amount, channel, merchant_gmv	amount, channel, merchant_gmv	0.691	0.0364	0.255

Table 3: WHICH accounts. Own-ref clock. Three readouts on the same slice: MMD ($P(X)$), CMean ($\|\Delta\mathbb{E}[X]\|$ and $\Delta\mathbb{E}[Y]$), PO-risk ($P(Y | X)$). Subset key is region, not Y .

kind	grain	south own-ref				north own-ref			
		MMD	$\ \Delta X\ $	PO	$\Delta\mathbb{E}[Y]$	MMD	$\ \Delta X\ $	PO	$\Delta\mathbb{E}[Y]$
covariate	order	0.35	2.55	$7.59e-05$	0.349	0.00109	0.302	0.000898	-0.0489
covariate	merchant	0.291	1.7	0.000475	0.349	-0.00538	0.196	0.000359	-0.0489
covariate	user	-0.00133	0.0659	0.00329	0.349	-0.000738	0.189	0.000486	-0.0489
covariate	all	0.296	3.07	0.000224	0.349	0.000566	0.407	0.00116	-0.0489
concept	order	-0.00137	0.275	0.00307	-0.0412	0.00109	0.302	0.000898	-0.0489
concept	merchant	-0.00969	0.059	0.000936	-0.0412	-0.00538	0.196	0.000359	-0.0489
concept	user	-0.00133	0.0659	0.000132	-0.0412	-0.000738	0.189	0.000486	-0.0489
concept	all	-0.00413	0.289	0.00376	-0.0412	0.000566	0.407	0.00116	-0.0489
both	order	0.35	2.55	0.0015	-0.262	0.00109	0.302	0.000898	-0.0489
both	merchant	0.291	1.7	0.000157	-0.262	-0.00538	0.196	0.000359	-0.0489
both	user	-0.00133	0.0659	0.00183	-0.262	-0.000738	0.189	0.000486	-0.0489
both	all	0.296	3.07	0.000552	-0.262	0.000566	0.407	0.00116	-0.0489

Read-off. Order grain + ADDIS hits at the labeled onset ($d=0$). The concatenated **all** grain is FAR at $t=0$. User X does not move (MMD ≈ 0); a T mark there is Y walking through another grain. FSDS recovers the planted columns on the grain that actually contains them. Covariate south order: MMD 0.35, $\|\Delta X\| = 2.55$, PO = $7.59e-05$, $\Delta\mathbb{E}[Y] = 0.349$; north MMD = 0.00109. Concept south order: MMD = -0.00137, PO = 0.00307, $\Delta\mathbb{E}[Y] = -0.0412$.